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Course: **Java FSE Digital Nurture**

Week 1

Exercise 1: Singleton Pattern

Aim

To implement the Singleton Design Pattern in Java.

Theory

The Singleton Pattern is a Creational Design Pattern that ensures only one instance of a class exists and provides a global point of access to it.

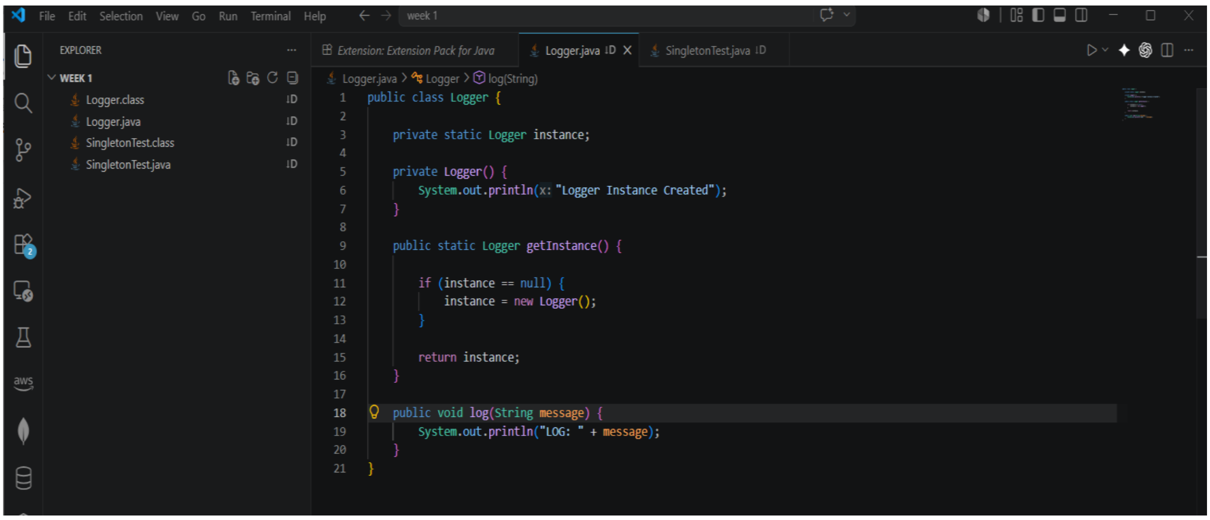
Algorithm

1. Create a class Logger.
2. Make its constructor private.
3. Create a private static instance variable.
4. Create a public static method getInstance().
5. Return the same instance whenever called.
6. Test using multiple references.

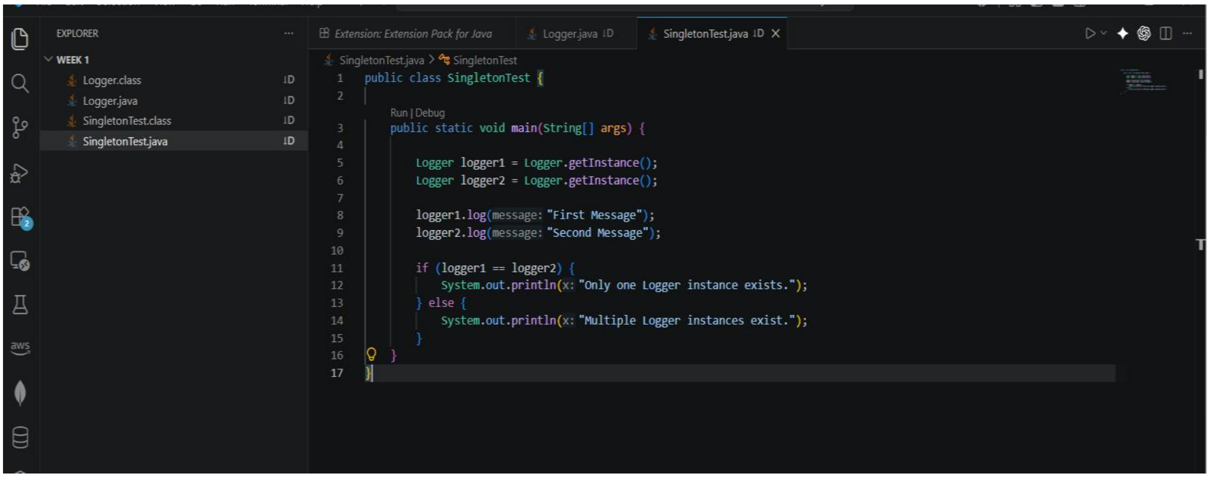
Program

Paste the full code for:

Logger.java



SingletonTest.java



OUTPUT:

Result

The Singleton Pattern was successfully implemented and verified.

Exercise 2: Implementing the Factory Method Pattern

Aim

To implement the Factory Method Design Pattern for creating different types of documents such as Word, PDF, and Excel documents.

Theory

The Factory Method Pattern is a Creational Design Pattern that provides an interface for creating objects while allowing subclasses or factory methods to determine the type of object to create. This pattern helps in achieving loose coupling between object creation and usage.

Algorithm

1. Create an interface named Document.
2. Define an open() method in the interface.
3. Create WordDocument, PdfDocument, and ExcelDocument classes implementing the Document interface.
4. Create a DocumentFactory class to generate different document objects.
5. Create a test class to demonstrate the creation of various document types.
6. Execute the program and verify the output.

Program

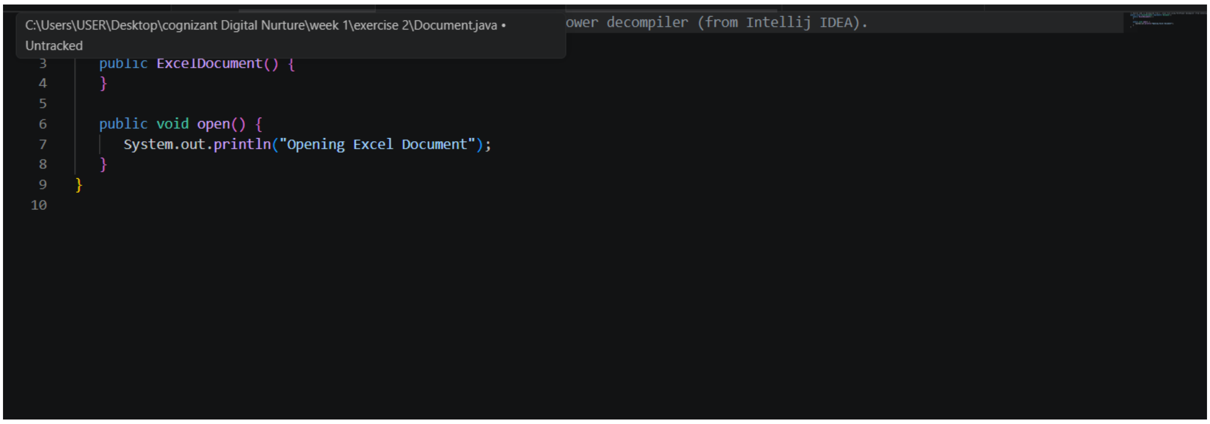
```java
// Source code is decompiled from a .class file using FernFlower decompiler (from Intellij IDEA).
public interface Document {
   void open();
}
```

exercise 2 > PdfDocument.java > PdfDocument

```java
public class PdfDocument implements Document {

    @Override
    public void open() {
        System.out.println(x: "Opening PDF Document");
    }
}
```

ExcelDocument.java



Factorymethodtest.java

Output